

# Apache JMeter

Supercharge Your QA Game with Speed,  
Stability & Scalability - Part 1

By Rashmi Panditharathne



# What is Apache JMeter?

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- Open-source performance testing tool by **Apache Foundation**
- Built using Java – runs on all platforms
- No licensing cost (Freeware)
- Competes with commercial tools via plugins and extensibility
- Strong community support + 3rd-party plugin ecosystem
- Supports coding in BeanShell, Groovy, JavaScript



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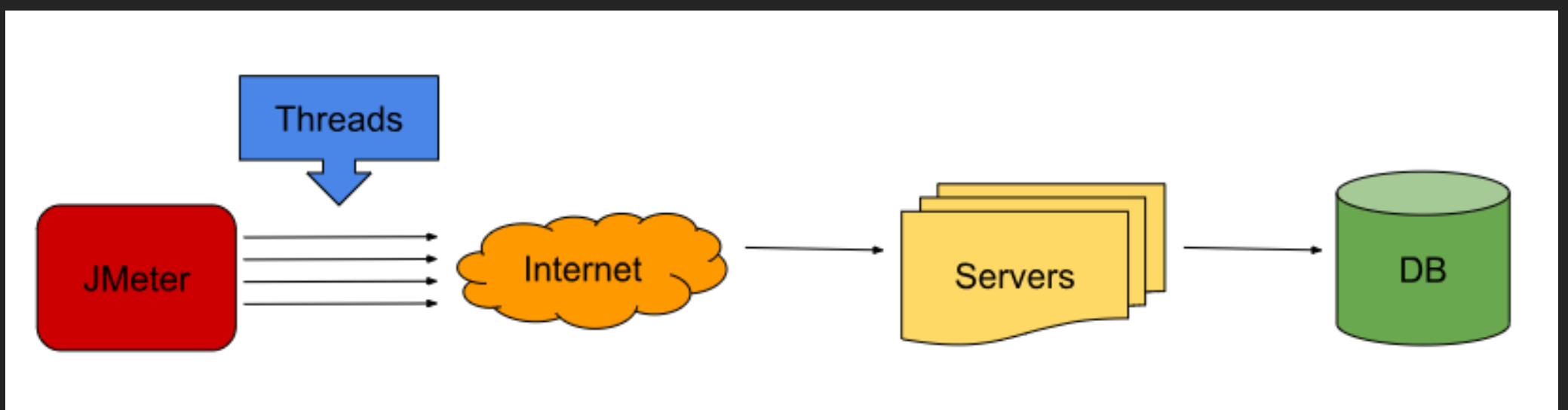


# How JMeter Works?

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JMeter simulates real-world users by creating Java threads that send requests over the internet to evaluate application performance.

- JMeter creates Java threads (virtual users)
- Threads simulate real users sending HTTP/API/DB requests
- Server processes these and responds
- JMeter captures & validates performance (response time, errors, etc.)



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# Key Features of JMeter

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- Platform-independent, **lightweight** & Open-source
- No license or maintenance cost
- Works on Windows, macOS, Linux
- Run in **GUI** and **Non-GUI** (headless) mode
- **Unlimited threads** (depends on system resources)
- Record & replay user actions
- Support for **mobile (Android), APIs, DBs, Web, SOAP, etc.**
- Script using **Groovy, Beanshell, JavaScript**
- CI/CD tools (e.g. **Jenkins**) integration
- Plugins support: Grafana, InfluxDB, Dynatrace

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# Limitations of JMeter

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- Limited native protocol support (e.g.no native SAP, Citrix)
- Needs plugins or coding for complex logic
- No real-time graphing (unless integrated with Grafana/InfluxDB)
- Memory-intensive on heavy loads
- Unpredictable behavior in distributed mode
- No "undo" or easy script regeneration
- Report analysis not as deep as tools like LoadRunner



# JMeter Installation

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## Prerequisites:

- Install Java (Check version: `java -version`)
- Add Java to `Environment Variables` → `System Variables` → `Path`

## Installation Steps:

1. Download JMeter from: [jmeter.apache.org](http://jmeter.apache.org)

2. Extract the ZIP (Binary version)

3. Launch:

- Windows → `/bin/jmeter.bat`
- macOS/Linux → `sh jmeter.sh`

✓ Tip: No installer needed – just extract & run.

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# JMeter Folder Structure

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## **apache-jmeter-<version>**

All files sit under this root directory

### 1. **/bin/** – Core execution & config

- **jmeter.bat, jmeter.sh** – GUI & Non-GUI launchers
- **jmeter.properties, system.properties** – Configs
- **log4j2.xml** – Logging configuration
- Subfolders:
  - **examples/** – Sample **.jmx** test plans
  - **templates/** – Ready-to-use test templates
  - **report-template/** – HTML report themes and layouts

### 2. **/lib/** – Libraries & Plugins

- Core JMeter JARs
- **/ext/** – External JARs (custom plugins)
- **/junit/** – JUnit support for unit testing



### 3. [/extras/](#)

- Tools like SSL certificate generators
- Optional scripts for advanced features

### 4. [/docs/](#)

- Offline HTML-based documentation

### 5. [/backups/](#)

- Auto-saved test plan backups (helps in recovery)

### 6. [/licenses/](#)

- Licenses of all third-party libraries used

### 7. [/printable\\_docs/](#)

- PDF versions of JMeter documentation

### 8. Other Files:

- [README.md](#), [LICENSE](#), [NOTICE](#) – About the tool





# How to Open JMeter (GUI Mode)

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## Option 1: Double-click Method (Windows):

- Navigate to `apache-jmeter-<version>/bin/`
- Run `jmeter.bat`

## Option 2: Via CMD:

### Windows:

```
cd C:\apache-jmeter-5.6.1\bin
jmeter.bat
```

### macOS/Linux:

```
cd ~/apache-jmeter-5.6.1/bin
sh jmeter.sh
```

 Use GUI for scripting | Non-GUI for execution in load tests

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# JMeter Core Elements

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## **Test Plan:**

Main container holding entire test flow

## **Thread Group (Users):**

Define virtual users, ramp-up, loop count

## **Samplers:**

Types of requests to send (e.g., HTTP Request, JDBC Request, FTP Request, SOAP/XML-RPC)

## **Config Elements:**

Set default values (e.g., CSV Data Set Config, HTTP Request Defaults)

## **Logic Controllers:**

Manage request execution flow (e.g., If, While, Loop, Switch)

## **Timers:**

Add delays (e.g., Constant Timer, Uniform Random Timer)

## **Assertions:**

Validate response content/status (e.g., Response Assertion, Duration Assertion)

## **Pre/Post Processors:**

Modify request/response before or after sampler execution

## **Listeners:**

View results: Table, Tree, Graph, Summary, HTML Report

## **Non-Test Elements:**

Debug Sampler, WorkBench (scratchpad)

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# Ready to Level Up Your QA Game?

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📎 Let's connect, share, and grow  
together.



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